Acids, bases and salts

Subject: Chemistry | Level: Class 10 | Curriculum: Federal Board
Prepared by: SHS Virtual Academy

1. Topic Overview and Learning Objectives

The study of acids, bases, and salts is the cornerstone of inorganic chemistry and is essential for understanding how substances interact in our daily lives. From the digestion of food in our stomachs to the cleaning agents we use at home, these chemical species govern a vast array of biological and industrial processes. Mastering this topic is not just about memorizing definitions; it is about grasping the fundamental nature of chemical reactivity and equilibrium.

For your Federal Board examinations, this chapter carries significant weight, often appearing in both the objective and subjective sections of the paper. Examiners frequently test your ability to differentiate between various theories, such as Arrhenius, Bronsted-Lowry, and Lewis concepts. Understanding the pH scale and the properties of salts is crucial for solving numerical problems and explaining chemical phenomena, making this chapter a high-yield area for scoring top marks.

The intuition behind this topic lies in the movement of protons and electrons. Whether a substance donates a hydrogen ion or accepts an electron pair, these interactions define the acidic or basic character of a solution. By focusing on the underlying mechanisms rather than just rote memorization, you will develop a strong foundation that will serve you well in higher-level chemistry studies and practical laboratory work.

2. Core Concepts and Detailed Revision Notes

2.1 Theories of Acids and Bases

To understand acids and bases, we look at three major historical theories. The Arrhenius concept defines acids as substances that provide H+ ions in aqueous solutions, while bases provide OH- ions. However, this theory is limited to aqueous environments. The Bronsted-Lowry theory expanded this by defining acids as proton donors and bases as proton acceptors, which works in non-aqueous solvents as well. Finally, the Lewis concept is the most comprehensive, defining acids as electron pair acceptors and bases as electron pair donors. This allows us to classify substances that do not even contain hydrogen, such as boron trifluoride, as acids. Understanding these three perspectives is vital because board questions often ask you to classify a specific reaction based on one of these theories, so you must be able to identify the donor and acceptor in every chemical equation provided. It is important to remember that these theories are not mutually exclusive but rather represent an evolution in our understanding of chemical bonding and reactivity.

Key Points to Remember:

1. Arrhenius acids produce H+ ions in water.
2. Bronsted-Lowry theory focuses on proton transfer.
3. Lewis theory defines acids as electron pair acceptors.
4. Conjugate acid-base pairs are formed by the transfer of a proton.
5. Amphoteric substances can act as both acids and bases.

2.2 The pH Scale and Indicators

The pH scale is a logarithmic measure used to express the concentration of hydrogen ions in a solution. It ranges from 0 to 14, where 7 is neutral, values below 7 indicate acidity, and values above 7 indicate basicity. Because the scale is logarithmic, a change of one pH unit represents a tenfold change in hydrogen ion concentration, which is a concept frequently tested in numerical problems. Indicators are weak organic acids or bases that change color depending on the pH of the solution. Common indicators like phenolphthalein and methyl orange are used in titrations to determine the equivalence point. When you are solving problems involving pH, remember that pH plus pOH always equals 14 at room temperature. This relationship allows you to easily calculate the basicity of a solution if you know its acidity, provided you keep your units consistent throughout your calculations. Always ensure your calculator is set correctly when performing log operations.

Key Points to Remember:

1. pH = -log(H+ concentration).
2. pOH = -log(OH- concentration).
3. pH + pOH = 14 at 25 degrees Celsius.
4. Indicators show color changes at specific pH ranges.
5. Universal indicator provides a full spectrum of pH values.

2.3 Properties and Preparation of Salts

Salts are ionic compounds formed by the neutralization reaction between an acid and a base. They consist of a positive ion from the base and a negative ion from the acid. Salts are generally crystalline solids with high melting points and are often soluble in water. There are several ways to prepare salts, including the reaction of an acid with a metal, a metal oxide, or a metal carbonate. You should be familiar with the difference between normal salts, acidic salts, and basic salts. Acidic salts are formed by partial replacement of replaceable hydrogen ions, while basic salts are formed by partial replacement of hydroxyl groups. In the examination, you might be asked to write balanced chemical equations for the preparation of specific salts, so ensure you practice your stoichiometry and balancing skills regularly to avoid losing marks on simple errors. Remember that the solubility of salts varies, and understanding these trends is helpful for practical chemistry.

Key Points to Remember:

1. Salts are formed by neutralization reactions.
2. Normal salts contain no replaceable H+ or OH- ions.
3. Acidic salts are formed by partial neutralization of polyprotic acids.
4. Salts can be prepared by direct combination or double displacement.
5. Crystallization is a common method for salt purification.

2.4 Neutralization and Titration

Neutralization is the reaction between an acid and a base to produce salt and water. This process is fundamental to analytical chemistry, particularly in titrations. A titration is a laboratory technique where a solution of known concentration is used to determine the concentration of an unknown solution. By carefully adding the titrant until the indicator changes color, we can calculate the exact molarity of the analyte. For board exams, you must understand the concept of the equivalence point, where the moles of H+ ions equal the moles of OH- ions. Always remember to use the titration formula M1V1/n1 = M2V2/n2 when performing calculations for titrations, where n represents the stoichiometry coefficients from the balanced equation. Being precise with your volume measurements and understanding the stoichiometry of the balanced equation is the key to getting the correct answer in these types of practical-based problems. Practice balancing equations to ensure you identify the correct n values.

Key Points to Remember:

1. Neutralization reaction: Acid + Base -> Salt + Water.
2. Titration determines unknown concentration using a standard solution.
3. Equivalence point is where the reaction is stoichiometrically complete.
4. End point is where the indicator changes color.
5. Molarity calculations are essential for titration problems.

3. Key Definitions, Formulas, and Rules

1. **pH definition:** pH = -log(H+)
2. **pOH definition:** pOH = -log(OH-)
3. **pH and pOH relationship:** pH + pOH = 14
4. **Water dissociation constant:** Kw = (H+)(OH-) = 10^-14
5. **Neutralization reaction:** Acid + Base -> Salt + Water
6. **Molarity formula:** M = moles of solute / liters of solution
7. **Titration formula:** M1V1/n1 = M2V2/n2
8. **Arrhenius Acid:** Substance that increases H+ concentration in water
9. **Lewis Base:** Substance that donates an electron pair

4. Step-by-Step Worked Examples

Example 1: Calculating pH from Molarity

Problem: Calculate the pH of a 0.01 M solution of HCl.

Solution Steps:

Step 1: Identify the concentration of H+ ions: HCl is a strong acid, so (H+) = 0.01 M.

Step 2: Use the pH formula: pH = -log(H+).

Step 3: Substitute the value: pH = -log(0.01).

Step 4: Calculate the result: pH = -(-2) = 2.

Final Answer: pH = 2

Example 2: Finding pOH

Problem: If the pH of a solution is 4, what is the pOH?

Solution Steps:

Step 1: Use the relationship formula: pH + pOH = 14.

Step 2: Substitute the known value: 4 + pOH = 14.

Step 3: Solve for pOH: pOH = 14 - 4.

Step 4: Calculate the result: pOH = 10.

Final Answer: pOH = 10

Example 3: Neutralization Stoichiometry

Problem: How many moles of NaOH are required to neutralize 0.5 moles of H2SO4?

Solution Steps:

Step 1: Write the balanced equation: H2SO4 + 2NaOH -> Na2SO4 + 2H2O.

Step 2: Identify the mole ratio: 1 mole of H2SO4 reacts with 2 moles of NaOH.

Step 3: Calculate: 0.5 moles H2SO4 - 2 = 1.0 mole NaOH.

Final Answer: 1.0 mole of NaOH

Example 4: Titration Calculation

Problem: 20 ml of 0.1 M NaOH is used to neutralize 10 ml of HCl. Find the molarity of HCl.

Solution Steps:

Step 1: Use the titration formula: M1V1 = M2V2 (since ratio is 1:1).

Step 2: Set M1 = 0.1 M, V1 = 20 ml, V2 = 10 ml.

Step 3: Solve for M2: M2 = (0.1 - 20) / 10.

Step 4: Calculate: M2 = 2 / 10 = 0.2 M.

Final Answer: 0.2 M

5. Common Exam Mistakes to Avoid

1. Forgetting to convert volume to liters when calculating molarity.
2. Confusing the pH and pOH scales; always check which one you are calculating.
3. Neglecting the stoichiometry of the balanced equation during neutralization problems.
4. Assuming all acids have only one replaceable hydrogen ion.
5. Writing incorrect chemical formulas for common salts like sodium carbonate.
6. Failing to include the negative sign in the logarithmic pH calculation.

6. Practice Questions and Model Answers

Question 1

Define a Lewis acid with one example.

Answer 1: A Lewis acid is a substance that can accept an electron pair. Example: BF3.

Question 2

Why is water considered amphoteric?

Answer 2: Water is amphoteric because it can act as both an acid (donating a proton) and a base (accepting a proton) depending on the reaction.

Question 3

Calculate the pH of a 0.001 M NaOH solution.

Answer 3: pOH = -log(0.001) = 3. pH = 14 - 3 = 11.

Question 4

What is the difference between a strong and weak acid?

Answer 4: A strong acid completely ionizes in water, whereas a weak acid only partially ionizes.

Question 5

Name the salt formed by the reaction of HCl and KOH.

Answer 5: The salt formed is Potassium Chloride (KCl).